Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/09/2010
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rohit.k.trivedi@gmail.com	**Fax:**	FY 515-294-4291
PI Organization Type:	UNIVERSITY	**Phone:**	515-450-1948
Organization Name:	Iowa State University		
PI Address 1:	Dept. of Materials Science and Engineering		
PI Address 2:	100 Wilhelm Hall		
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011	**Congressional District:**	4
Comments:	Other email: trivedi@ameslab.gov		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 5/31/2012 (from 5/31/2010) per NSSC info dtd 9/2009 (10/2009) NOTE: Received NCE to 5/31/2010 (from 5/31/2009) per NSSC info dtd 6/2009 (10/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	The proposed research will consist of accurate and rigorous numerical modeling of the dynamics of pattern evolution, coupled with critical ground-based and low gravity experimental studies in a model transparent system in which the dynamics of pattern evolution can be observed in situ. To obtain experimental data under diffusive growth conditions, ground-based experiments were carried out in thin samples. Two different configurations were used: a thin rectangular sample in which only one array of cells or dendrites forms, and a cylindrical capillary sample in which only a single cell or dendrite forms. These experiments gave a quantitative understanding of cell and dendrite formation in one and two dimensions. All experiments were carried out in the succinonitrile-0.25 wt% camphor to establish the range of conditions over which cellular and dendrtic structures are stable under diffusive growth conditions. Specific aspects that were quantitatively examined include the conditions for the cell-dendrite transition and the variation in the shape of the cell, tip radius, spacing and tip temperature as a function of velocity and thermal gradient. Since only one composition is used in the space experiment, the composition effect of the pattern evolution was examined in some ground-based experiments to isolate the effect of composition from those of velocity and thermal gradient.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict and design reliable processing conditions that optimize microstructures and hence material properties and performance. These results are crucial in improving castings for aerospace and defense industries. In addition, the knowledge gained from this study will be applied to other pattern forming phenomena that occur in different fields of science such as combustion, fluid physics, cryo-biology, and food science. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions.
Task Progress:	A detailed set of experiments was carried out in which the microstructure variation was examined over a wide range of velocities that give rise to planar, cellular and dendritic microstructures, and all experiments were carried out in succininitrile-0.25 wt % camphor, which is the alloy composition that is used in the space experiments in DSI unit in DECLIC facility. These results were critical in determining the experimental matrix that we are now using for experiments on the space station. Ground-based experiments were carried out for two different configurations of thin samples in which convection effects are negligible so that critical information required for the flight experimental matrix can be determined. In addition, they provide quantitative information on pattern formation in one and two dimensions that can be quantitatively analyzed. Experiments were carried out in thin rectangular samples in which a two-dimensional array of cells or dendrites form, and in a cylindrical tube of capillary dimension in which a single cell or dendrite forms. Both these techniques give diffusive growth conditions so that the results can be useful for the design of the flight experiments. Experiments were completed to characterize the conditions of velocity and thermal gradient that give rise to cellular structures and dendritic structure and the combination of these two structures. Detailed measurements were made to establish the conditions for the cell to dendrite transition, and it was found that the transition depends on the local spacing of cells. In addition, it is found that the primary spacing undergoes a discontinuous increase as the transition from cell to dendrite occurs. The variation in cell and dendrite spacing, tip radius and tip temperature were quantitatively characterized as a function of velocity and thermal gradient. Since the precise thermal gradient in space experiment is not known, different values of thermal gradients were used so that the data can be used for flight experiment once the gradient is determined in DSI-DECLIC experiment. Detailed modeling of the interface pattern evolution was developed by using the phase-field approach. Simulations were carried out for a single cell or dendrite in a cylindrical geometry with varying diameter of the tube and for an array of cells in two dimensions with varying thickness of the sample. We also examined the formation of three-dimensional array with hexagonal and square arrangement of cells and dendrites. Finally, simulations were carried out to observe the dynamical evolution of spatial pattern in three-dimensions from the initial planar interface to the final cellular array. It was shown that the anisotropy in interface energy is crucial in stabilizing steady-state growth, and the value of the anisotropy parameter is crucial in predicting the scale of the pattern. These calculations will now be continued for the succinionitrile-0.25 wt% camphor so that the model can be used to analyze the results to be obtained from the DSI-DECLIC experiments on the International Space Station.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Asta M, Beckermann C, Karma A, Kurz W, Napolitano R, Plapp M, Purdy G, Rappaz M, Trivedi R. "Solidification microstructures and solid-state parallels: Recent developments, future directions." Acta Materialia 2009 Feb;57(4):941-71. http://dx.doi.org/10.1016/j.actamat.2008.10.020 , Feb-2009
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Articles in Peer-reviewed Journals	Gurevich S, Karma A, Plapp M, Trivedi R. "Phase-field study of three-dimensional steady-state growth shapes in directional solidification." Physical Review E 2010 Jan;81(1):011603 (15 p.). http://dx.doi.org/10.1103/PhysRevE.81.011603 , Jan-2010
Books/Book Chapters	Trivedi R, Sunseri E. "Non-Plane Front Solidification." in "ASM Handbook, vol. 15, Casting." Ed. S. Viswanathan et al. Materials Park, Ohio : ASM International, 2008. p. 299-306., Dec-2008